

MACHINE LEARNING

Teaching computers to learn patterns from data — without programming every rule.

DATA → **LEARNING** → **PREDICTION**

Examples of patterns a model can learn:

- Which emails look like spam
- What a handwritten digit represents
- How a house price relates to its features

CORE IDEA

**Learn from examples.
Find useful relationships.
Make predictions on new
data.**

THREE MAIN TYPES OF ML

The difference is how much feedback the model receives while learning.

01

Supervised

Labeled data: inputs with known answers

Spam detection · Price prediction

02

Unsupervised

Unlabeled data: discover hidden structure

Customer segments · Anomaly detection

03

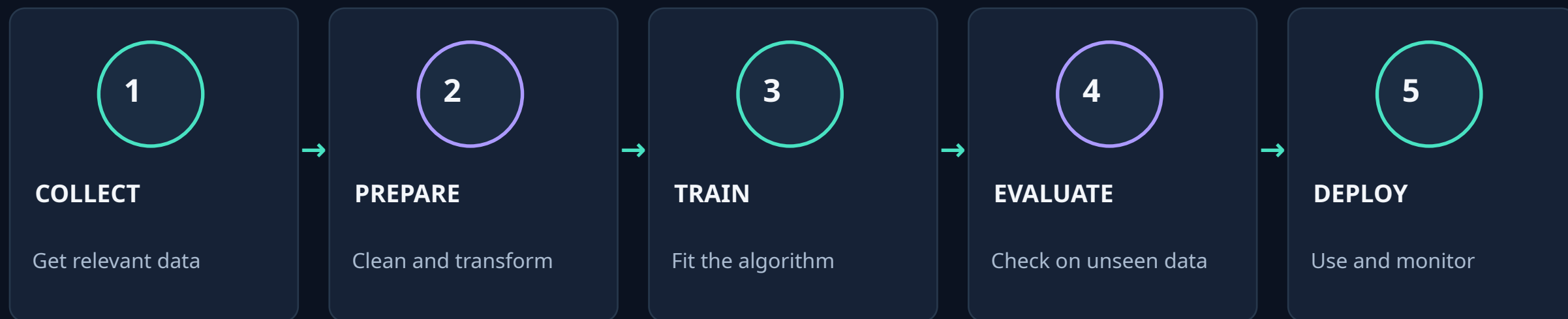
Reinforcement

An agent learns via actions and rewards

Game playing · Robotics

HOW A MACHINE LEARNING PROJECT WORKS

A repeatable workflow turns raw data into useful predictions.



Important: separate training and test data to measure generalization.

POPULAR ALGORITHMS

Different tools fit different kinds of problems.

Linear Regression

Predict a number

House prices

Decision Tree / Random Forest

Classify or predict

Fraud detection

K-Means

Find groups

Customer segments

Neural Networks

Learn complex patterns

Images and language

Algorithm selection depends on the task, dataset size, interpretability and evaluation results.

WHERE MACHINE LEARNING IS USED

Machine learning powers products and decisions across industries.

HEALTHCARE

Assisting diagnosis and risk prediction

FINANCE

Detecting suspicious transactions

RETAIL

Personalized recommendations

TRANSPORT

Predicting traffic and optimizing routes

TAKEAWAY **Good ML = quality data + suitable models + careful evaluation + responsible use.**